

MySQL BR Conf 2026

Por que fazer upgrade do seu MySQL

e novidades da versão 9



BIOGRAFIAS

Diego Silva Martins



Consultor DBA e Tech Lead



LINKEDIN

[linkedin.com/in/dsm82](https://www.linkedin.com/in/dsm82)



INSTAGRAM

[@_diegosmartins](https://www.instagram.com/_diegosmartins)



BIOGRAFIAS

Andryos Luigi Ribeiro



LINKEDIN

linkedin.com/in/andryosr



INSTAGRAM

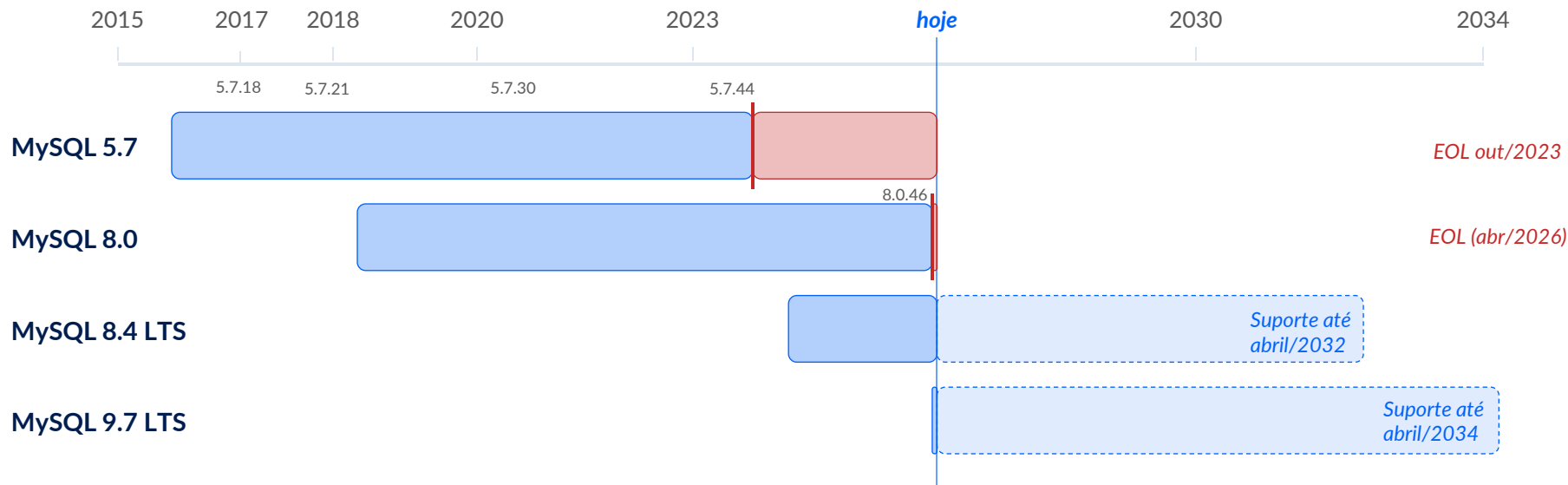
[@andryosr](https://instagram.com/andryosr)



CONTEXTO

Onde estamos hoje

Linha do tempo do suporte oficial das versões MySQL



Apenas duas versões recebem patches hoje: **8.4 LTS e 9.7 LTS**. Toda outra versão é, tecnicamente, vulnerável.

O peso da segurança

Cada trimestre sem upgrade é um trimestre de exposição acumulada.



3+

anos

que o MySQL 5.7 está sem patches de segurança



30+

correções

por trimestre na Critical Patch Update da Oracle



250+

CVEs

publicados em MySQL desde o EOL do 5.7 (out/2023)

*Cada trimestre, a Oracle libera em média mais de 30 correções de segurança para MySQL. **Para quem está na versão 5.7, são 3 anos sem atualizações.***

Top 5 CVEs versão 5.7

7.5

CVSS

CVE-2024-20973 Server: Optimizer - Falha no componente Optimizer que permite a um usuário autenticado de baixo privilégio induzir consumo descontrolado de recursos via queries com subqueries, JOINS ou hints de otimização especialmente estruturados. O mysqld entra em loop ou esgota CPU, resultando em hang ou crash de repetição.

6.5

CVSS

CVE-2024-20961 Server: Optimizer - Qualquer usuário autenticado no banco pode enviar uma consulta especialmente formulada que trava o servidor completamente. Não precisa de privilégios elevados — uma conta de aplicação comum já basta

6.5

CVSS

CVE-2024-20985 Server: UDF (User Defined Functions) - Funções personalizadas do banco (UDFs) podem ser acionadas de forma maliciosa por qualquer usuário autenticado, incluindo contas de serviço de sistemas integrados — que costumam ter credenciais fracas e nunca são auditada

6.3

CVSS

CVE-2024-20964 Server: Security - Um usuário com acesso limitado consegue, explorando essa falha, obter permissões além do que foi concedido a ele. Na prática, uma conta de leitura pode passar a modificar ou apagar dados de áreas que não deveria tocar

5.5

CVSS

CVE-2024-20967 Server: Replication - A vulnerabilidade atinge a camada de replicação e pode gerar divergência de dados entre o servidor principal e suas réplicas, sem gerar alertas visíveis.

Estes são apenas 5 dos **250+ CVEs publicados desde o EOL do MySQL 5.7.**

Além da segurança

Quatro razões adicionais que pesam tanto quanto CVEs e patches.



Compliance

LGPD, PCI-DSS e ISO 27001 exigem software com suporte ativo. Auditorias reprovam versões EOL, principalmente na área financeira.



Cloud providers

AWS RDS encerrou suporte padrão ao 5.7 em fev/2024. Azure descontinuou. Continuar custa mais ou simplesmente não é opção.



Risco operacional

Sem suporte oficial, qualquer incidente vira crítico, e o pior: sem correção. Não há para onde escalar.



Custo oculto

Equipe gasta horas contornando limitações antigas. Cada workaround é dívida técnica que cresce com juros.

O QUE VOCÊ GANHA

Ganhos técnicos: do 5.7 ao 8.4

Mais que segurança – performance e recursos que impactam no negócio



Performance

até 3x

mais TPS sob alta concorrência

- 8.4 mantém TPS estável a 512 threads onde 5.7 colapsa
- Hash Joins: queries analíticas várias vezes mais rápidas
- Histogramas automáticos: o otimizador escolhe o plano certo sozinho

Recursos novos (5.7 → 8.4)

• Window Functions

• CTEs (Common Table Expressions)

• Roles (RBAC)

• Instant ADD COLUMN

• Atomic DDL

• innodb_redo_log_capacity (8.4)

• Invisible & Descending indexes

Como fazer o upgrade

Cinco etapas para execução do upgrade com segurança e assertividade

01

Assessment

Inventário completo:
versão, drivers, queries,
jobs e integrações.

02

Análise

checkForServerUpgrade,
pt-upgrade, sql_mode,
charset e deprecations.

03

Plano

Estratégia 5.7→8.0→8.4
via réplica + cutover.
Janela curta.

04

Execução

Upgrade com replicação,
validação contínua e plano
de rollback ativo.

05

Pós-upgrade

Tuning, observabilidade,
ajuste de planos e
acompanhamento
dedicado.



Desafio nº 1: caching_sha2_password

Plugin de autenticação mudou no 8.0 e foi desabilitado por padrão no 8.4. Drivers PHP, JDBC, ODBC, .NET e sistemas legados (Delphi, VB6) precisam ser verificados e atualizados antes. O upgrade não é só do servidor – é também das pontas.

Principais validações para o upgrade

MySQL Shell - `util.checkForServerUpgrade()` – Execução mandatória



Estrutura de tabelas

- Tabelas sem chave primária (PK)
- MyISAM no schema mysql
- FK's em colunas que não são PK's



Palavras reservadas

- Reserved words como nomes
- GROUP BY com ASC/DESC implícito



Charset e collation

- Uso de utf8mb3 (deprecated)
- Collations removidas ou depreciadas



Dados e tipos

- Datas/horas zero (0000-00-00)
- Tipos de coluna obsoletos



Configuração e variáveis

- Variáveis de sistema removidas
- Defaults alterados (sql_mode etc.)
- Compatibilidade de configuração



Plugins e privilégios

- mysql_native_password obsoleto
- DEFINERs inexistentes em views
- Privilégios SUPER → granulares

UPGRADE - Hands ON

```
mysql> select @@version;
+-----+
| @@version |
+-----+
| 5.7.44-log |
+-----+
1 row in set (0.00 sec)

mysql> show databases;
+-----+
| Database |
+-----+
| information_schema |
| loja_demo |
| mysql |
| performance_schema |
| sys |
+-----+
5 rows in set (0.01 sec)

mysql> █
```

UPGRADE - Hands ON

```
[ec2-user@ip-172-31-1-254 ~]$ mysqlsh root@localhost:3306
MySQL Shell 8.4.10
```

Copyright (c) 2016, 2026, Oracle and/or its affiliates.
Oracle is a registered trademark of Oracle Corporation and/or its affiliates.
Other names may be trademarks of their respective owners.

```
Type '\help' or '\?' for help; '\quit' to exit.
Creating a Classic session to 'root@localhost:3306'
Fetching global names for auto-completion... Press ^C to stop.
Your MySQL connection id is 16
Server version: 5.7.44-log MySQL Community Server (GPL)
No default schema selected; type \use <schema> to set one.
MySQL localhost:3306 ssl SQL > \js
Switching to JavaScript mode...
```

```
MySQL localhost:3306 ssl JS > util.checkForServerUpgrade("root@localhost:3306", {targetVersion: "8.4.8",
configPath: "/etc/my.cnf"})
```

```
Please provide the password for 'root@localhost:3306': *****
Save password for 'root@localhost:3306'? [Y]es/[N]o/[N]ever (default No): yes
The MySQL server at localhost:3306, version 5.7.44-log - MySQL Community Server
(GPL), will now be checked for compatibility issues for upgrade to MySQL 8.4.8.
```

WARNING: Upgrading MySQL Server from version 5.7.44 to 8.4.8 is not supported.
Please consider running the check using the following option: targetVersion=8.0

```
1) Usage of old temporal type (oldTemporal)
   No issues found
```

UPGRADE - Hands ON

2) MySQL syntax check for routine-like objects (routineSyntax)
No issues found

3) Usage of db objects with names conflicting with new reserved keywords (reservedKeywords)

Warning: The following objects have names that conflict with new reserved keywords. Ensure queries sent by your applications use `quotes` when referring to them or they will result in errors.

loja_demo.produtos.rank - Column name
loja_demo.vw_ranking_produtos.rank - Column name

More information:

<https://dev.mysql.com/doc/refman/en/keywords.html>

4) Usage of utf8mb3 charset (utf8mb3)

Warning: The following objects use the deprecated utf8mb3 character set. It is recommended to convert them to use utf8mb4 instead, for improved Unicode support. The utf8mb3 character is subject to removal in the future.

loja_demo - schema's default character set: utf8
loja_demo.categorias.codigo - column's default character set: utf8
loja_demo.categorias.nome - column's default character set: utf8
loja_demo.clientes.nome - column's default character set: utf8

UPGRADE - Hands ON

```
loja_demo.clientes.uf - column's default character set: utf8
loja_demo.log_acessos.usuario - column's default character set: utf8
loja_demo.log_acessos.acao - column's default character set: utf8
loja_demo.lojas.nome - column's default character set: utf8
loja_demo.lojas.cidade - column's default character set: utf8
loja_demo.pedidos.status - column's default character set: utf8
loja_demo.produtos.nome - column's default character set: utf8
loja_demo.produtos.categoria_codigo - column's default character set: utf8
loja_demo.vw_pedidos_detalhados.status - column's default character set: utf8
loja_demo.vw_pedidos_detalhados.cliente - column's default character set:
utf8
loja_demo.vw_pedidos_detalhados.produto - column's default character set:
utf8
loja_demo.vw_ranking_produtos.nome - column's default character set: utf8
loja_demo.vw_ranking_produtos.categoria - column's default character set:
utf8
loja_demo.vw_vendas_por_cliente.nome - column's default character set: utf8
loja_demo.vw_vendas_por_cliente.cidade - column's default character set: utf8
```

More information:

<https://dev.mysql.com/doc/refman/en/charset-unicode-utf8mb3.html>

5) Table names in the mysql schema conflicting with new tables in the latest MySQL. (mysqlSchema)
No issues found

UPGRADE - Hands ON

- 6) Partitioned tables using engines with non native partitioning
(nonNativePartitioning)

Error: In the latest MySQL storage engine is responsible for providing its own partitioning handler, and the MySQL server no longer provides generic partitioning support. InnoDB and NDB are the only storage engines that provide a native partitioning handler that is supported in the latest MySQL. A partitioned table using any other storage engine must be altered—either to convert it to InnoDB or NDB, or to remove its partitioning—before upgrading the server, else it cannot be used afterwards.

loja_demo.log_acessos - MyISAM engine does not support native partitioning

More information:

<https://dev.mysql.com/doc/refman/8.0/en/upgrading-from-previous-series.html#upgrade-configuration-changes>

- 7) Foreign key constraint names longer than 64 characters (foreignKeyLength)
No issues found
- 8) Usage of obsolete MAXDB sql_mode flag (maxdbSqlModeFlags)
No issues found

UPGRADE - Hands ON

9) Usage of obsolete sql_mode flags (obsoleteSqlModeFlags)

Notice: The following DB objects have obsolete options persisted for sql_mode, which will be cleared during the upgrade.

```
loja_demo.fn_coordenadas_loja - FUNCTION uses obsolete NO_AUTO_CREATE_USER
sql_mode
loja_demo.fn_hash_senha_legacy - FUNCTION uses obsolete NO_AUTO_CREATE_USER
sql_mode
loja_demo.fn_total_pedido - FUNCTION uses obsolete NO_AUTO_CREATE_USER
sql_mode
loja_demo.sp_gerar_clientes - PROCEDURE uses obsolete NO_AUTO_CREATE_USER
sql_mode
loja_demo.sp_gerar_pedidos - PROCEDURE uses obsolete NO_AUTO_CREATE_USER
sql_mode
loja_demo.sp_relatorio_mensal - PROCEDURE uses obsolete NO_AUTO_CREATE_USER
sql_mode
loja_demo.trg_itens_pedido_ai - TRIGGER uses obsolete NO_AUTO_CREATE_USER
sql_mode
@@global.sql_mode - defined using obsolete NO_AUTO_CREATE_USER option
```

More information:

<https://dev.mysql.com/doc/refman/8.0/en/mysql-nutshell.html#mysql-nutshell-removals>

10) ENUM/SET column definitions containing elements longer than 255 characters (enumSetElementLength)

No issues found

UPGRADE - Hands ON

11) Usage of partitioned tables in shared tablespaces

(partitionedTablesInSharedTablespaces)

No issues found

12) Circular directory references in tablespace data file paths

(circularDirectory)

No issues found

13) Usage of removed functions (removedFunctions)

Error: The following DB objects use functions that were removed in the latest MySQL version. Please make sure to update them to use supported alternatives before upgrade.

loja_demo.fn_coordenadas_loja - FUNCTION uses removed functions Y (consider using ST_Y instead), X (consider using ST_X instead)
loja_demo.fn_hash_senha_legado - FUNCTION uses removed function PASSWORD

More information:

<https://dev.mysql.com/doc/refman/8.0/en/mysql-nutshell.html#mysql-nutshell-removals>

14) Usage of removed GROUP BY ASC/DESC syntax (groupbyAscSyntax)

Error: The following DB objects use removed GROUP BY ASC/DESC syntax. They need to be altered so that ASC/DESC keyword is removed from GROUP BY clause and placed in appropriate ORDER BY clause.

loja_demo.sp_relatorio_mensal - PROCEDURE uses removed GROUP BY ASC syntax

UPGRADE - Hands ON

More information:

<https://dev.mysql.com/doc/relnotes/mysql/8.0/en/news-8-0-13.html#mysqld-8-0-13-sql-syntax>

15) Removed system variables for error logging to the system log configuration
(removedSysLogVars)

No issues found

16) Removed system variables (removedSysVars)

Error: Following system variables that were detected as being used will be removed. Please update your system to not rely on them before the upgrade.

expire_logs_days - Error: The system variable 'expire_logs_days' is set to 7 (SERVER) and will be removed.

query_cache_size - Error: The system variable 'query_cache_size' is set to 16M (SERVER) and will be removed.

query_cache_type - Error: The system variable 'query_cache_type' is set to 1 (SERVER) and will be removed.

More information:

<https://dev.mysql.com/doc/refman/8.0/en/added-deprecated-removed.html#optvars-removed>

UPGRADE - Hands ON

17) System variables with new default values (sysVarsNewDefaults)

Warning: Following system variables that are not defined in your configuration file will have new default values. Please review if you rely on their current values and if so define them before performing upgrade.

back_log - default value will change.

collation_server - default value will change from latin1_swedish_ci to utf8mb4_0900_ai_ci.

event_scheduler - default value will change from OFF to ON.

explicit_defaults_for_timestamp - default value will change from OFF to ON.

innodb_autoinc_lock_mode - default value will change from 1 (consecutive) to 2 (interleaved).

innodb_flush_method - default value will change from NULL to fsync (Unix), unbuffered (Windows).

innodb_flush_neighbors - default value will change from 1 (enable) to 0 (disable).

innodb_max_dirty_pages_pct - default value will change from 75 (%) 90 (%).

innodb_max_dirty_pages_pct_lwm - default value will change from 0 (%) to 10 (%).

innodb_undo_log_truncate - default value will change from OFF to ON.

innodb_undo_tablespace - default value will change from 0 to 2.

log_error_verbosity - default value will change from 3 (Notes) to 2 (Warning).

log_slave_updates - default value will change from OFF to ON.

master_info_repository - default value will change from FILE to TABLE.

max_allowed_packet - default value will change from 4194304 (4MB) to 67108864 (64MB).

UPGRADE - Hands ON

max_error_count - default value will change from 64 to 1024.
optimizer_trace_max_mem_size - default value will change from 16KB to 1MB.
performance_schema_consumer_events_transactions_current - default value will change from OFF to ON.
performance_schema_consumer_events_transactions_history - default value will change from OFF to ON.
relay_log_info_repository - default value will change from FILE to TABLE.
slave_rows_search_algorithms - default value will change from 'INDEX_SCAN, TABLE_SCAN' to 'INDEX_SCAN, HASH_SCAN'.
table_open_cache - default value will change from 2000 to 4000.
transaction_write_set_extraction - default value will change from OFF to XXHASH64.
binlog_transaction_dependency_tracking - default value will change from COMMIT_ORDER to WRITESET.
group_replication_consistency - default value will change from EVENTUAL to BEFORE_ON_PRIMARY_FAILOVER.
group_replication_exit_state_action - default value will change from READ_ONLY to OFFLINE_MODE.
innodb_adaptive_hash_index - default value will change from ON to OFF.
innodb_buffer_pool_in_core_file - default value will change from ON to OFF.
innodb_buffer_pool_instances - default value will change from 8 (or 1 if innodb_buffer_pool_size < 1GB) to MAX(1, #vcpu/4).
innodb_change_buffering - default value will change from all to none.
innodb_doublewrite_files - default value will change from innodb_buffer_pool_instances * 2 to 2.
innodb_doublewrite_pages - default value will change from innodb_write_io_threads to 128.

UPGRADE - Hands ON

`innodb_flush_method` - default value will change from `fsynch (unix)` or `unbuffered (windows)` to `O_DIRECT`.
`innodb_io_capacity` - default value will change from 200 to 10000.
`innodb_io_capacity_max` - default value will change from 200 to 2 x `innodb_io_capacity`.
`innodb_log_buffer_size` - default value will change from 16777216 (16MB) to 67108864 (64MB).
`innodb_log_writer_threads` - default value will change from ON to OFF (if `#vcpu <= 32`).
`innodb_numa_interleave` - default value will change from OFF to ON.
`innodb_page_cleaners` - default value will change from 4 to `innodb_buffer_pool_instances`.
`innodb_parallel_read_threads` - default value will change from 4 to `MAX(#vcpu/8, 4)`.
`innodb_purge_threads` - default value will change from 4 to 1 (if `#vcpu <= 16`).
`innodb_read_io_threads` - default value will change from 4 to `MAX(#vcpu/2, 4)`.
`innodb_redo_log_capacity` - default value will change from 104857600 (100MB) to `MIN ( #vcpu/2, 16 )GB`.

More information:

<https://dev.mysql.com/blog-archive/new-defaults-in-mysql-8-0/>

UPGRADE - Hands ON

18) Zero Date, Datetime, and Timestamp values (zeroDates)

Warning: By default zero date/datetime/timestamp values are no longer allowed in MySQL, as of 5.7.8 NO_ZERO_IN_DATE and NO_ZERO_DATE are included in SQL_MODE by default. These modes should be used with strict mode as they will be merged with strict mode in a future release. If you do not include these modes in your SQL_MODE setting, you are able to insert date/datetime/timestamp values that contain zeros. It is strongly advised to replace zero values with valid ones, as they may not work correctly in the future.

loja_demo.clientes.criado_em - column has zero default value: 0000-00-00 00:00:00

More information:

<https://dev.mysql.com/doc/refman/8.0/en/datetime.html>

https://dev.mysql.com/doc/refman/8.0/en/sql-mode.html#sqlmode_no_zero_date

https://dev.mysql.com/doc/refman/8.0/en/sql-mode.html#sqlmode_no_zero_in_date

19) Schema inconsistencies resulting from file removal or corruption

(schemaInconsistency)

No issues found

20) Tables recognized by InnoDB that belong to a different engine (engineMixup)

No issues found

21) Issues reported by 'check table x for upgrade' command (checkTableCommand)

No issues found

UPGRADE - Hands ON

22) New default authentication plugin considerations
(defaultAuthenticationPlugin)

Warning: The default authentication plugin 'caching_sha2_password' offers more secure password hashing than previously used 'mysql_native_password' (and consequent improved client connection authentication). However, it also has compatibility implications that may affect existing MySQL installations. If your MySQL installation must serve pre-8.0 clients and you encounter compatibility issues after upgrading, the simplest way to address those issues is to reconfigure the server to revert to the previous default authentication plugin (mysql_native_password). For example, use these lines in the server option file:

```
[mysqld]  
default_authentication_plugin=mysql_native_password
```

However, the setting should be viewed as temporary, not as a long term or permanent solution, because it causes new accounts created with the setting in effect to forego the improved authentication security.

MySQL 8.4.0 removes the deprecated `default_authentication_plugin` option. The deprecated `mysql_native_password` authentication plugin is disabled by default as of MySQL 8.4.0, and is subject to removal in a future version.

If you are using replication please take time to understand how the authentication plugin changes may impact you.

UPGRADE - Hands ON

More information:

<https://dev.mysql.com/doc/refman/8.0/en/upgrading-from-previous-series.html#upgrade-caching-sha2-password-compatibility-issues>

<https://dev.mysql.com/doc/refman/8.0/en/upgrading-from-previous-series.html#upgrade-caching-sha2-password-replication>

<https://dev.mysql.com/doc/refman/8.4/en/mysql-nutshell.html>

23) Indexes on functions with changed semantics

(changedFunctionsInGeneratedColumns)

No issues found

24) Columns which cannot have default values (columnsWhichCannotHaveDefaults)

No issues found

25) Check for invalid table names and schema names used in 5.7 (invalid57Names)

No issues found

26) Check for orphaned routines and events in 5.7 (orphanedObjects)

No issues found

27) Check for deprecated usage of single dollar signs in object names

(dollarSignName)

No issues found

28) Check for indexes that are too large to work on higher versions of MySQL Server than 5.7 (indexTooLarge)

No issues found

UPGRADE - Hands ON

29) Check for deprecated '`<table>`' syntax used in routines.

(emptyDotTableSyntax)

No issues found

30) Check for columns that have foreign keys pointing to tables from a different database engine. (invalidEngineForeignKey)

No issues found

31) Checks for foreign keys not referencing a full unique index

(foreignKeyReferences)

Foreign keys to partial indexes may be forbidden as of 8.4.0, this check identifies such cases to warn the user.

**loja_demo.fk_produtos_categoria - invalid foreign key defined as
'produtos(categoria_codigo)' references a non unique key at table
'categorias'.**

Solutions:

- Convert non unique key to unique key if values do not have any duplicates.
In case of foreign keys involving partial columns of key, create composite unique key containing all the referencing columns if values do not have any duplicates.
- Remove foreign keys referring to non unique key/partial columns of key.
- In case of multi level references which involves more than two tables change foreign key reference.

UPGRADE - Hands ON

32) Check for deprecated or invalid user authentication methods.

(authMethodUsage)

Some users are using authentication methods that may be deprecated or removed, please review the details below.

Warning: The following users are using the 'mysql_native_password' authentication method which is deprecated as of MySQL 8.0.34 and will be removed in a future release.

Consider switching the users to a different authentication method (i.e. caching_sha2_password).

The 'mysql_native_password' authentication type is disabled by default in MySQL 8.4, but can still be enabled by setting

loose_mysql_native_password=ON.

- app_loja@%
- mysql.session@localhost
- mysql.sys@localhost
- relatorio@localhost
- root@localhost

More information:

<https://dev.mysql.com/doc/refman/en/caching-sha2-pluggable-authentication.html>

33) Check for deprecated or removed plugin usage. (pluginUsage)

No issues found

UPGRADE - Hands ON

34) Check for deprecated or invalid default authentication methods in system variables. (deprecatedDefaultAuth)

The following variables have problems with their set authentication method:

```
Error: default_authentication_plugin - mysql_native_password authentication
method was removed and it must be corrected before upgrading to 8.4.0
release.
```

35) Check for deprecated or invalid authentication methods in use by MySQL Router internal accounts. (deprecatedRouterAuthMethod)

No issues found

36) Check for deprecated temporal delimiters in table partitions. (deprecatedTemporalDelimiter)

No issues found

37) Checks for errors in column definitions (columnDefinition)

No issues found

38) Check for allowed values in System Variables. (sysvarAllowedValues)

No issues found

39) Checks for partitions by key using columns with prefix key indexes (partitionsWithPrefixKeys)

No issues found

UPGRADE - Hands ON

Errors: 8

Warnings: 74

Notices: 8

ERROR: 8 errors were found. Please correct these issues before upgrading to avoid compatibility issues.

MySQL localhost:3306 ssl JS >

UPGRADE - Hands ON

```
-- =====
-- 02_correcoes_para_80.sql - Correções apontadas pelo checker (5.7 -> 8.0)
-- Rodar AINDA NO 5.7: mysql -uroot -p < 02_correcoes_para_80.sql
-- =====

USE loja_demo;

-- Rotinas recriadas aqui ficam sem NO_AUTO_CREATE_USER
SET SESSION sql_mode = 'ONLY_FULL_GROUP_BY,STRICT_TRANS_TABLES,NO_ZERO_IN_DATE,NO_ZERO_DATE,ERROR_FOR_DIVISION_E

-----
-- [ERRO] Particionamento não nativo: MyISAM particionado -> InnoDB
-----

ALTER TABLE log_acessos ENGINE = InnoDB;

-----
-- [ERRO] Funções removidas: PASSWORD(), X(), Y()
-----

DROP FUNCTION IF EXISTS fn_hash_senha_legacy;
DROP FUNCTION IF EXISTS fn_coordenadas_loja;

DELIMITER $$

CREATE FUNCTION fn_hash_senha_legacy(p_senha VARCHAR(100))
RETURNS VARCHAR(64)
DETERMINISTIC NO SQL
BEGIN
    RETURN SHA2(p_senha, 256);
END$$

CREATE FUNCTION fn_coordenadas_loja(p_loja_id INT)
RETURNS VARCHAR(60)
READS SQL DATA
BEGIN
    DECLARE v_coord VARCHAR(60);
    SELECT CONCAT(ST_Y(localizacao), ', ', ST_X(localizacao))
    INTO v_coord
    FROM lojas
    WHERE id = p_loja_id;
    RETURN v_coord;
END$$
```

UPGRADE - Hands ON

32) Check for deprecated or removed plugin usage. (pluginUsage)

No issues found

33) Check for deprecated or invalid default authentication methods in system variables. (deprecatedDefaultAuth)

The following variables have problems with their set authentication method:

Warning: default_authentication_plugin - mysql_native_password
authentication method is deprecated and it should be considered to correct
this before upgrading to 8.4.0 release.

34) Check for deprecated or invalid authentication methods in use by MySQL Router internal accounts. (deprecatedRouterAuthMethod)

No issues found

35) Check for deprecated temporal delimiters in table partitions.
(deprecatedTemporalDelimiter)

No issues found

Errors: 0

Warnings: 30

Notices: 5

NOTE: No fatal errors were found that would prevent an upgrade, but some potential issues were detected.
Please ensure that the reported issues are not significant before upgrading.

MySQL localhost:3306 ssl JS >

UPGRADE - Hands ON

```
[ec2-user@ip-172-31-1-254 ~]$ sudo yum update -y mysql-community-server
Loaded plugins: amazon-id, search-disabled-repos
mysql80-ol7
mysql84-tools-ol7
(1/2): mysql80-ol7/x86_64/updateinfo
(2/2): mysql80-ol7/x86_64/primary_db
Resolving Dependencies
--> Running transaction check
--> Package mysql-community-server.x86_64 0:5.7.44-1.el7 will be updated
--> Package mysql-community-server.x86_64 0:8.0.46-1.el7 will be an update
--> Processing Dependency: mysql-community-common(x86-64) = 8.0.46-1.el7 for package: mysql-community-server-8.0.4
--> Processing Dependency: mysql-community-icu-data-files = 8.0.46-1.el7 for package: mysql-community-server-8.0.4
--> Processing Dependency: mysql-community-client(x86-64) >= 8.0.11 for package: mysql-community-server-8.0.46-1.e
--> Running transaction check
--> Package mysql-community-client.x86_64 0:5.7.44-1.el7 will be updated
--> Package mysql-community-client.x86_64 0:8.0.46-1.el7 will be an update
--> Processing Dependency: mysql-community-client-plugins = 8.0.46-1.el7 for package: mysql-community-client-8.0.4
--> Processing Dependency: mysql-community-libs(x86-64) >= 8.0.11 for package: mysql-community-client-8.0.46-1.el7
--> Package mysql-community-common.x86_64 0:5.7.44-1.el7 will be updated
--> Package mysql-community-common.x86_64 0:8.0.46-1.el7 will be an update
--> Package mysql-community-icu-data-files.x86_64 0:8.0.46-1.el7 will be installed
--> Running transaction check
--> Package mysql-community-client-plugins.x86_64 0:8.0.46-1.el7 will be installed
--> Package mysql-community-libs.x86_64 0:5.7.44-1.el7 will be updated
--> Package mysql-community-libs.x86_64 0:8.0.46-1.el7 will be an update
--> Finished Dependency Resolution

Dependencies Resolved
```

UPGRADE - Hands ON

```
[ec2-user@ip-172-31-1-254 ~]$ sudo service mysqld status
Redirecting to /bin/systemctl status mysqld.service
● mysqld.service - MySQL Server
   Loaded: loaded (/usr/lib/systemd/system/mysqld.service; enabled; vendor preset: disabled)
   Active: active (running) since Thu 2026-09-24 14:56:34 UTC; 3s ago
     Docs: man:mysqld(8)
           http://dev.mysql.com/doc/refman/en/using-systemd.html
  Process: 2238 ExecStartPre=/usr/bin/mysqld_pre_systemd (code=exited, status=0/SUCCESS)
 Main PID: 2266 (mysqld)
    Status: "Server is operational"
   CGroup: /system.slice/mysqld.service
           └─2266 /usr/sbin/mysqld

Sep 24 14:56:23 ip-172-31-1-254.sa-east-1.compute.internal systemd[1]: Starting MySQL Server...
Sep 24 14:56:34 ip-172-31-1-254.sa-east-1.compute.internal systemd[1]: Started MySQL Server.
[ec2-user@ip-172-31-1-254 ~]$ mysql
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 11
Server version: 8.0.46 MySQL Community Server - GPL

Copyright (c) 2000, 2026, Oracle and/or its affiliates.

Oracle is a registered trademark of Oracle Corporation and/or its
affiliates. Other names may be trademarks of their respective
owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> select @@version;
+-----+
| @@version |
+-----+
| 8.0.46    |
+-----+
1 row in set (0.00 sec)
```

UPGRADE - Hands ON

```
ec2-user@ip-172-31-1-254 ~]$ sudo grep -i upgrade /var/log/mysqld.log | tail -5
2026-09-24T14:56:23.881339Z 1 [System] [MY-011012] [Server] Starting upgrade of data directory.
2026-09-24T14:56:27.240387Z 5 [System] [MY-013381] [Server] Server upgrade from '50700' to '80046' started.
2026-09-24T14:56:34.583002Z 5 [System] [MY-013381] [Server] Server upgrade from '50700' to '80046' completed.
ec2-user@ip-172-31-1-254 ~]$
ec2-user@ip-172-31-1-254 ~]$
ec2-user@ip-172-31-1-254 ~]$
```

UPGRADE - Hands ON

```
[ec2-user@ip-172-31-1-254 ~]$ mysqlsh root@localhost:3306
MySQL Shell 8.4.10

Copyright (c) 2016, 2026, Oracle and/or its affiliates.
Oracle is a registered trademark of Oracle Corporation and/or its affiliates.
Other names may be trademarks of their respective owners.

Type '\help' or '\?' for help; '\quit' to exit.
Creating a Classic session to 'root@localhost:3306'
Fetching global names for auto-completion... Press ^C to stop.
Your MySQL connection id is 8
Server version: 8.0.46 MySQL Community Server - GPL
No default schema selected; type \use <schema> to set one.
MySQL localhost:3306 ssl SQL > \js
Switching to JavaScript mode...
MySQL localhost:3306 ssl JS > util.checkForServerUpgrade("root@localhost:3306", {targetVersion: "8.4.8", configPath: "/etc/my.cnf"})
The MySQL server at localhost:3306, version 8.0.46 - MySQL Community Server -
GPL, will now be checked for compatibility issues for upgrade to MySQL 8.4.8.

1) Removed system variables (removedSysVars)
   No issues found
```

UPGRADE - Hands ON

```
ensure they stop being used before the upgrade as they will be lost.
```



```
12) Checks for partitions by key using columns with prefix key indexes
(partitionsWithPrefixKeys)
    No issues found
Errors:    0
Warnings: 22
Notices:  2
```



```
NOTE: No fatal errors were found that would prevent an upgrade, but some potential issues were detected. Ple
```



```
MySQL localhost:3306 ssl JS > \quit
Bye!
[ec2-user@ip-172-31-1-254 ~]$
```

UPGRADE - Hands ON

```
Updating:
  mysql-community-server                                x86_64                                8.4.
Updating for dependencies:
  mysql-community-common                                x86_64                                8.4.
  mysql-community-icu-data-files                        x86_64                                8.4.

Transaction Summary
=====
Upgrade 1 Package (+2 Dependent packages)

Total download size: 64 M
Downloading packages:
Delta RPMs disabled because /usr/bin/applydeltarpm not installed.
(1/3): mysql-community-icu-data-files-8.4.10-1.el7.x86_64.rpm
(2/3): mysql-community-common-8.4.10-1.el7.x86_64.rpm
(3/3): mysql-community-server-8.4.10-1.el7.x86_64.rpm
-----
Total
Running transaction check
Running transaction test
Transaction test succeeded
Running transaction
  Updating : mysql-community-icu-data-files-8.4.10-1.el7.x86_64
  Updating : mysql-community-common-8.4.10-1.el7.x86_64
  Updating : mysql-community-server-8.4.10-1.el7.x86_64
warning: /etc/my.cnf created as /etc/my.cnf.rpmnew
  Cleanup  : mysql-community-server-8.0.46-1.el7.x86_64
  Cleanup  : mysql-community-common-8.0.46-1.el7.x86_64
  Cleanup  : mysql-community-icu-data-files-8.0.46-1.el7.x86_64
  Verifying : mysql-community-common-8.4.10-1.el7.x86_64
  Verifying : mysql-community-server-8.4.10-1.el7.x86_64
  Verifying : mysql-community-icu-data-files-8.4.10-1.el7.x86_64
  Verifying : mysql-community-common-8.0.46-1.el7.x86_64
  Verifying : mysql-community-server-8.0.46-1.el7.x86_64
  Verifying : mysql-community-icu-data-files-8.0.46-1.el7.x86_64

Updated:
  mysql-community-server.x86_64 0:8.4.10-1.el7

Dependency Updated:
  mysql-community-common.x86_64 0:8.4.10-1.el7                                mysql-com

Complete!
```

UPGRADE - Hands ON

```
Updating:
  mysql-community-server                                x86_64                                8.4.
Updating for dependencies:
  mysql-community-common                                x86_64                                8.4.
  mysql-community-icu-data-files                        x86_64                                8.4.

Transaction Summary
=====
Upgrade 1 Package (+2 Dependent packages)

Total download size: 64 M
Downloading packages:
Delta RPMs disabled because /usr/bin/applydeltarpm not installed.
(1/3): mysql-community-icu-data-files-8.4.10-1.el7.x86_64.rpm
(2/3): mysql-community-common-8.4.10-1.el7.x86_64.rpm
(3/3): mysql-community-server-8.4.10-1.el7.x86_64.rpm
-----
Total
Running transaction check
Running transaction test
Transaction test succeeded
Running transaction
  Updating : mysql-community-icu-data-files-8.4.10-1.el7.x86_64
  Updating : mysql-community-common-8.4.10-1.el7.x86_64
  Updating : mysql-community-server-8.4.10-1.el7.x86_64
warning: /etc/my.cnf created as /etc/my.cnf.rpmnew
  Cleanup  : mysql-community-server-8.0.46-1.el7.x86_64
  Cleanup  : mysql-community-common-8.0.46-1.el7.x86_64
  Cleanup  : mysql-community-icu-data-files-8.0.46-1.el7.x86_64
  Verifying : mysql-community-common-8.4.10-1.el7.x86_64
  Verifying : mysql-community-server-8.4.10-1.el7.x86_64
  Verifying : mysql-community-icu-data-files-8.4.10-1.el7.x86_64
  Verifying : mysql-community-common-8.0.46-1.el7.x86_64
  Verifying : mysql-community-server-8.0.46-1.el7.x86_64
  Verifying : mysql-community-icu-data-files-8.0.46-1.el7.x86_64

Updated:
  mysql-community-server.x86_64 0:8.4.10-1.el7

Dependency Updated:
  mysql-community-common.x86_64 0:8.4.10-1.el7                                mysql-com

Complete!
```

UPGRADE - Hands ON

```
[ec2-user@ip-172-31-1-254 ~]$  
[ec2-user@ip-172-31-1-254 ~]$ sudo service mysqld start  
Redirecting to /bin/systemctl start mysqld.service  
  
[ec2-user@ip-172-31-1-254 ~]$  
[ec2-user@ip-172-31-1-254 ~]$ sudo grep -i upgrade /var/log/mysqld.log | tail -5  
2026-09-24T14:56:27.240387Z 5 [System] [MY-013381] [Server] Server upgrade from '50700' to '80046' started.  
2026-09-24T14:56:34.583002Z 5 [System] [MY-013381] [Server] Server upgrade from '50700' to '80046' completed.  
2026-09-24T16:56:50.065028Z 1 [System] [MY-013413] [Server] Data dictionary upgrade from version '80023' to '80300' completed.  
2026-09-24T16:56:51.855654Z 4 [System] [MY-013381] [Server] Server upgrade from '80046' to '80410' started.  
2026-09-24T16:56:57.699880Z 4 [System] [MY-013381] [Server] Server upgrade from '80046' to '80410' completed.  
[ec2-user@ip-172-31-1-254 ~]$  
[ec2-user@ip-172-31-1-254 ~]$  
[ec2-user@ip-172-31-1-254 ~]$
```

UPGRADE - Hands ON

```
[ec2-user@ip-172-31-1-254 ~]$ mysql -uapp_loja -p'L0ja#Demo2026' -h127.0.0.1 loja_demo -e "SELECT CURRENT_USER(), VERSION();"
mysql: [Warning] Using a password on the command line interface can be insecure.
+-----+-----+
| CURRENT_USER() | VERSION() |
+-----+-----+
| app_loja@%     | 8.4.10    |
+-----+-----+
[ec2-user@ip-172-31-1-254 ~]$
[ec2-user@ip-172-31-1-254 ~]$ mysql
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 13
Server version: 8.4.10 MySQL Community Server - GPL

Copyright (c) 2000, 2026, Oracle and/or its affiliates.

Oracle is a registered trademark of Oracle Corporation and/or its
affiliates. Other names may be trademarks of their respective
owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> select user,host,plugin from mysql.user;
+-----+-----+-----+
| user          | host      | plugin                               |
+-----+-----+-----+
| app_loja      | %         | caching_sha2_password              |
| mysql.infoschema | localhost | caching_sha2_password              |
| mysql.session | localhost | caching_sha2_password              |
| mysql.sys     | localhost | caching_sha2_password              |
| relatorio     | localhost | caching_sha2_password              |
| root          | localhost | caching_sha2_password              |
+-----+-----+-----+
6 rows in set (0.00 sec)
```

MySQL 9.7 — O que há de novo?

Lançada em 21/abr/2026. Vem amadurecendo desde julho/2024 em [versões Innovation](#) — agora estável e LTS



Hypergraph Optimizer

Community

Novo otimizador para queries com muitos JOINs. Antes exclusivo do HeatWave — agora disponível no Community.



JSON Duality

Community

Aplicação enxerga documentos JSON; o banco mantém modelo relacional. Os dois mundos na mesma base.



Dynamic Data Masking

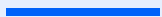
Enterprise

Policy aplicada na coluna da tabela base. Valor mascarado para quem não tem o role correto.

PARA TERMINAR



O custo de não fazer o upgrade **é sempre maior** do que o do próprio upgrade — só vem mais tarde, e cobrando juros.





“

Obrigado!